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June 21, 2026

To the Vercel Agent Stack Team,

An AI Gateway is a harder problem than it looks. It is not just routing — it is caching semantically equivalent prompts, enforcing rate limits under bursty agent traffic, failing over across providers without surfacing errors to developers, and doing all of it with the latency budget of a real-time application. I have built the routing and failover layer for a multi-LLM system and understand where these problems bite in production. I want to help build Vercel AI Gateway into the infrastructure that every TypeScript developer uses to talk to LLMs.

- › **Multi-Provider LLM Routing with Circuit Breakers: KubeServe** is a production LLM inference platform I built: FastAPI router with KEDA autoscaling (scale-to-zero to N replicas), multi-provider routing across Claude/GPT-4o/Gemini, circuit breaker + EWMA load balancing, and OTel + Prometheus + Grafana observability. Rate limiting, caching hot paths with Redis, and graceful provider failover are the core architectural problems I solved there.
- › **High-Throughput API Infrastructure in TypeScript:** At **Isomer AI**, I owned the full TypeScript codebase delivering 30+ customer deployments: Node.js + Express APIs, Redis caching, Auth0 RBAC, PostgreSQL, Playwright test coverage. In **AlgoPulse**, I built systems handling 2,000 async inference requests per second with sub-100ms leaderboard latency under k6 load tests at 10,000 concurrent users.
- › **AI/ML Integration and Observability:** Through KubeServe and Isomer, I integrated Claude, GPT-4o, and Gemini behind model-agnostic APIs with full OTel tracing and Prometheus metrics on every LLM call. The role calls out “AI/ML integrations, API gateways, distributed systems” — these are the three intersections where I have built production experience.
- › **Vercel Ecosystem Fluency: PulseRank** is a production event-driven system built in NestJS and Next.js — Vercel’s own flagship framework. I understand the developer experience Vercel optimizes for, and I want to build infrastructure that feels native to that ecosystem.

The AI Gateway is the connective tissue between every developer and every LLM on Vercel’s platform. I want to help build it right. I would be glad to discuss.

Sincerely,

Aditya Kulkarni